Footprint applications: Sustainability reporting

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Recap – Week 7

- Multi-region input-output tables can be generated using data from more than one region
- The regions may be different countries, or different regions within a particular country
- The diagonal of a MRIO represents domestic IO transactions and the off-diagonals represent trade matrices
- The location of import and export vectors in a MRIO table
- How MRIO tables are constructed, and how uncertainties are managed

Footprint applications: Sustainability reporting

How does input-output analysis help with sustainability reporting?

Sustainability reporting

Australian Sustainability Reporting Standard AASB S2

The Australian Accounting Standards Board makes Australian Sustainability Reporting Standard AASB S2 *Climate-related Disclosures* under section 336A of the *Corporations Act 2001*.

Keith Kendall
Chair – AASB

Dated 20 September 2024

Australian Sustainability Reporting Standard AASB S2
Climate-related Disclosures

Objective

1 **The objective of AASB S2 *Climate-related Disclosures* is to require an entity to disclose information about its *climate-related risks and opportunities* that is useful to *primary users of general purpose financial reports* in making decisions relating to providing resources to the entity.**[1]

2 This Standard requires an entity to disclose information about climate-related risks and opportunities that could reasonably be expected to affect the entity's cash flows, its access to finance or cost of capital over the short, medium or long term. For the purposes of this Standard, these risks and opportunities are collectively referred to as 'climate-related risks and opportunities that could reasonably be expected to affect the entity's prospects'.

Figure 1: TNFD's recommended disclosures

Governance	Strategy	Risk & impact management	Metrics & targets
Disclose the organisation's governance of nature-related dependencies, impacts, risks and opportunities.	Disclose the effects of nature-related dependencies, impacts, risks and opportunities on the organisation's business model, strategy and financial planning where such information is material.	Describe the processes used by the organisation to identify, assess, prioritise and monitor nature-related dependencies, impacts, risks and opportunities.	Disclose the metrics and targets used to assess and manage material nature-related dependencies, impacts, risks and opportunities.
Recommended disclosures	**Recommended disclosures**	**Recommended disclosures**	**Recommended disclosures**
A. Describe the board's oversight of nature-related dependencies, impacts, risks and opportunities. **B.** Describe management's role in assessing and managing nature-related dependencies, impacts, risks and opportunities. **C.** Describe the organisation's human rights policies and engagement activities, and oversight by the board and management, with respect to Indigenous Peoples, Local Communities, affected and other stakeholders, in the organisation's assessment of, and response to, nature-related dependencies, impacts, risks and opportunities.	**A.** Describe the nature-related dependencies, impacts, risks and opportunities the organisation has identified over the short, medium and long term. **B.** Describe the effect nature-related dependencies, impacts, risks and opportunities have had on the organisation's business model, value chain, strategy and financial planning, as well as any transition plans or analysis in place. **C.** Describe the resilience of the organisation's strategy to nature-related risks and opportunities, taking into consideration different scenarios. **D.** Disclose the locations of assets and/or activities in the organisation's direct operations and, where possible, upstream and downstream value chain(s) that meet the criteria for priority locations.	**A**(i) Describe the organisation's processes for identifying, assessing and prioritising nature-related dependencies, impacts, risks and opportunities in its direct operations. **A**(ii) Describe the organisation's processes for identifying, assessing and prioritising nature-related dependencies, impacts, risks and opportunities in its upstream and downstream value chain(s). **B.** Describe the organisation's processes for managing nature-related dependencies, impacts, risks and opportunities. **C.** Describe how processes for identifying, assessing, prioritising and monitoring nature-related risks are integrated into and inform the organisation's overall risk management processes.	**A.** Disclose the metrics used by the organisation to assess and manage material nature-related risks and opportunities in line with its strategy and risk management process. **B.** Disclose the metrics used by the organisation to assess and manage dependencies and impacts on nature. **C.** Describe the targets and goals used by the organisation to manage nature-related dependencies, impacts, risks and opportunities and its performance against these.

Figure 4
Recommendations and Supporting Recommended Disclosures

Governance	Strategy	Risk Management	Metrics and Targets
Disclose the organization's governance around climate-related risks and opportunities.	Disclose the actual and potential impacts of climate-related risks and opportunities on the organization's businesses, strategy, and financial planning where such information is material.	Disclose how the organization identifies, assesses, and manages climate-related risks.	Disclose the metrics and targets used to assess and manage relevant climate-related risks and opportunities where such information is material.
Recommended Disclosures	**Recommended Disclosures**	**Recommended Disclosures**	**Recommended Disclosures**
a) Describe the board's oversight of climate-related risks and opportunities.	a) Describe the climate-related risks and opportunities the organization has identified over the short, medium, and long term.	a) Describe the organization's processes for identifying and assessing climate-related risks.	a) Disclose the metrics used by the organization to assess climate-related risks and opportunities in line with its strategy and risk management process.
b) Describe management's role in assessing and managing climate-related risks and opportunities.	b) Describe the impact of climate-related risks and opportunities on the organization's businesses, strategy, and financial planning.	b) Describe the organization's processes for managing climate-related risks.	b) Disclose Scope 1, Scope 2, and, if appropriate, Scope 3 greenhouse gas (GHG) emissions, and the related risks.
	c) Describe the resilience of the organization's strategy, taking into consideration different climate-related scenarios, including a 2°C or lower scenario.	c) Describe how processes for identifying, assessing, and managing climate-related risks are integrated into the organization's overall risk management.	c) Describe the targets used by the organization to manage climate-related risks and opportunities and performance against targets.

AUSTRALIA

Modern Slavery Act 2018

No. 153, 2018

Compilation No. 2

Compilation date: 7 November 2024

Includes amendments: Act No. 42, 2024

TCFD: https://assets.bbhub.io/company/sites/60/2021/10/FINAL-2017-TCFD-Report.pdf
TNFD: https://tnfd.global/publication/recommendations-of-the-taskforce-on-nature-related-financial-disclosures/
Modern Slavery Act: https://www.legislation.gov.au/C2018A00153/latest/text
AASB Standard: https://standards.aasb.gov.au/aasb-s2-sep-2024

What information is needed?

b) Disclose Scope ~~1, Scope 2~~, and, if appropriate, Scope 3 greenhouse gas (GHG) emissions, and the related risks

TCFD

Climate-related metrics

29 An entity shall disclose information relevant to the cross-industry metric categories of:

(a) *greenhouse gases*—the entity shall:

(i) disclose its absolute gross greenhouse gas emissions generated during the reporting period, expressed as metric tonnes of CO_2 *equivalent* (see paragraphs B19–B22), classified as:

(1) *Scope 1 greenhouse gas emissions*;

(2) *Scope 2 greenhouse gas emissions*; and

(3) *Scope 3 greenhouse gas emissions*;

AASB S2

16 Mandatory criteria for modern slavery statements

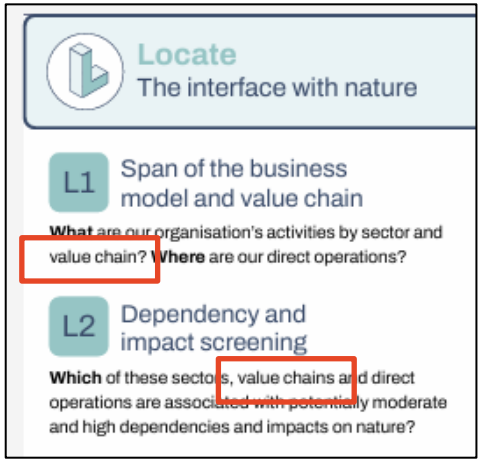
(1) A modern slavery statement must, in relation to each reporting entity covered by the statement:

(a) identify the reporting entity; and

(b) describe the structure, operations and supply chains of the reporting entity; and

(c) describe the ~~risks of modern~~ slavery practices in the operations and supply chains of the reporting entity, and any entities that the reporting entity owns or controls; and

Modern Slavery Act



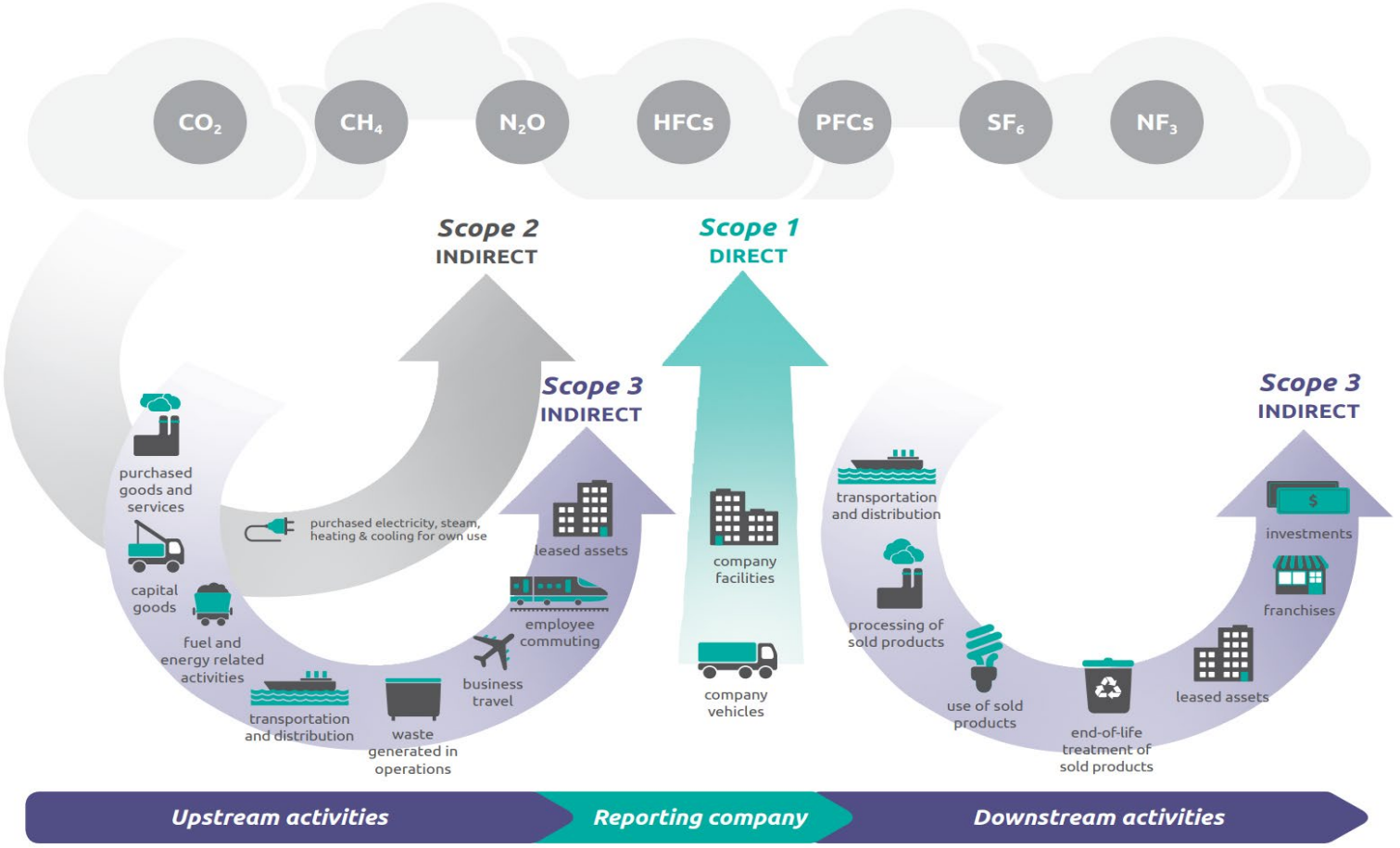
TNFD

Climate-related reporting

Scope 1	Direct operational emissions (e.g. on-site gas use, fuel for company cars, or on-site equipment)
Scope 2	Emissions as a result of the entity's purchase or acquisition of electricity
Scope 3	All other indirect emissions

Greenhouse gas protocol

Figure [I] Overview of GHG Protocol scopes and emissions across the value chain



Source: Figure 1.1 of *Scope 3 Standard*.

https://ghgprotocol.org/sites/default/files/standards/Scope3_Calculation_Guidance_0.pdf

Calculating Scope 1

Scope 1 emissions can be calculated using emissions factors and information on the amount of fuel used.

Note the units carefully – we need to do two conversions here:

1. Volume -> GJ of energy
2. GJ of energy -> CO_2-e

Table 9 Direct (scope 1) and indirect (scope 3) emission factors for the consumption of transport fuels in different transport equipment

Transport type	Fuel combusted	Energy Content Factor (GJ per unit of fuel)	Scope 1 Emission Factor (kg CO_2-e/GJ) CO_2	Scope 1 Emission Factor (kg CO_2-e/GJ) CH_4	Scope 1 Emission Factor (kg CO_2-e/GJ) N_2O	Scope 1 Emission Factor (kg CO_2-e/GJ) Combined gases	Scope 3 Emission Factor (kg CO_2-e/GJ)
Cars and light commercial vehicles	Gasoline	34.2	67.4	0.02*	0.2*	67.62	17.2
Cars and light commercial vehicles	Diesel oil 柴油	38.6	69.9	0.01*	0.5*	70.41	17.3
Cars and light commercial vehicles	Liquefied petroleum gas (LPG)	26.2	60.2	0.5*	0.3*	61.00	20.2
Cars and light commercial vehicles	Fuel oil	39.7	73.6	0.08	0.5	74.18	18.0
Cars and light commercial vehicles	Ethanol	23.4	0.0	0.2*	0.2*	0.4	NE
Cars and light commercial vehicles	Biodiesel	34.6	0.0	0.8	1.7	2.5	NE
Cars and light commercial vehicles	Renewable diesel	38.6	0.0	0.01*	0.5*	0.51	NE

https://www.dcceew.gov.au/climate-change/publications/national-greenhouse-accounts-factors-2026

Calculating Scope 2

Scope 2 emissions can be calculated using emissions factors and information on kWh of electricity purchased.

Table 1 Indirect (scope 2 and scope 3) location-based emission factors associated with the consumption and losses of purchased or acquired electricity

State, Territory or grid description	Scope 2 Emission Factors (kg CO_2-e/kWh)	Scope 3 Emission Factors (kg CO_2-e/kWh)
New South Wales and Australian Capital Territory	0.60	0.07
Victoria	0.74	0.11
Queensland	0.65	0.11
South Australia	0.21	0.03
Western Australia - South West Interconnected System (SWIS)	0.45	0.10
Western Australia - North Western Interconnected System (NWIS)	0.56	0.09
Tasmania	0.23	0.03
Northern territory - Darwin Katherine Interconnected System (DKIS)	0.55	0.13
National	0.59	0.10

Sources: Primary data sources comprise National Greenhouse and Energy Reporting (Measurement) Determination 2008 (Schedule 1), Australian Energy Statistics, Clean Energy Regulator, and AEMO data and Department of Climate Change, Energy, the Environment and Water.

https://www.dcceew.gov.au/climate-change/publications/national-greenhouse-accounts-factors-2026

Scope 3 emissions

The GHG Protocol suggests that entities report scope 3 emissions in 15 different categories:

1. Purchased goods & services
2. Capital goods
3. Fuel- and energy-related activities
4. Upstream transportation & distribution
5. Waste generated in operations
6. Business travel
7. Employee commuting
8. Upstream leased assets

Upstream emissions

9. Downstream transportation & distribution
10. Processing of sold products
11. Use of sold products
12. End-of-life treatment of sold products
13. Downstream leased assets
14. Franchises
15. Investments

Downstream emissions

https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf
https://ghgprotocol.org/sites/default/files/standards/Scope3_Calculation_Guidance_0.pdf

Defining the Scope 3 boundary

- Accounting for scope 3 emissions is optional under the GHG Protocol.
- Each entity defines which other entities it will include in its scope 3 calculations.
- Entities usually calculate these by asking their suppliers to provide information on their scope 1 & 2 emissions.

BOX 2. Organizational and operational boundaries

Organization X is a parent company that has full ownership and financial control of operations A and B, but only a 30% non-operated interest and no financial control in operation C.

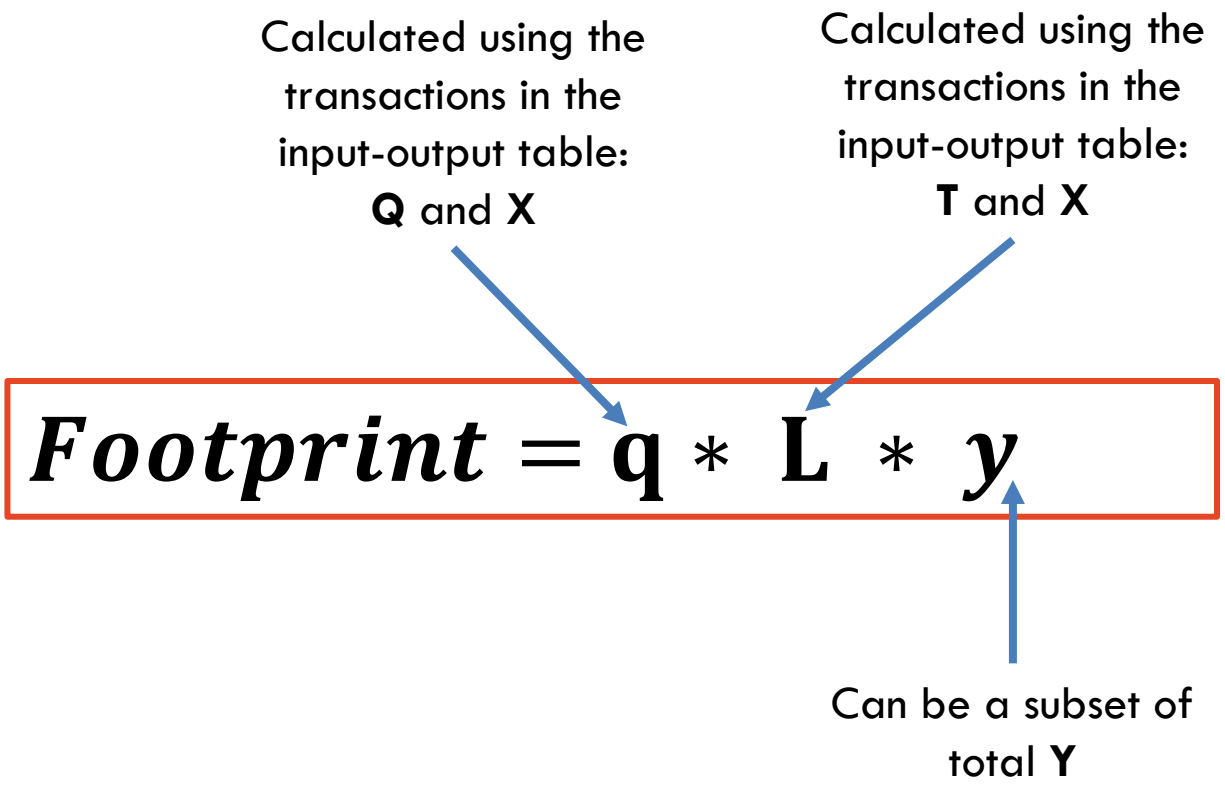
Setting Organizational Boundary: X would decide whether to account for GHG emissions by equity share or financial control. If the choice is equity share, X would include A and B, as well as 30% of C's emissions. If the approach chosen is financial control, X would count only A and B's emissions as relevant and subject to consolidation. Once this has been decided, the organizational boundary has been defined.

Setting Operational Boundary: Once the organizational boundary is set, X then needs to decide, on the basis of its business goals, whether to account only for scope 1 and scope 2, or whether to include relevant scope 3 categories for its operations.

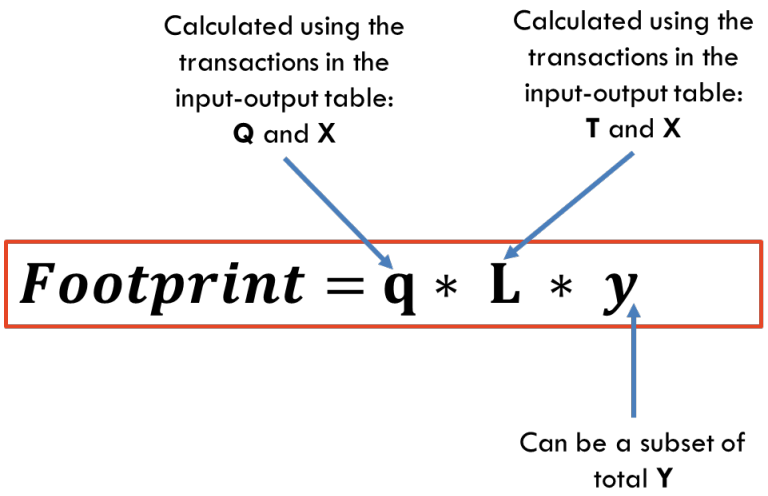
Operations A, B and C (if the equity approach is selected) account for the GHG emissions in the scopes chosen by X, i.e., they apply the corporate policy in drawing up their operational boundaries.

https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf

Calculating Scope 3 – part I



Calculating footprints for a particular entity



Note that these do not change

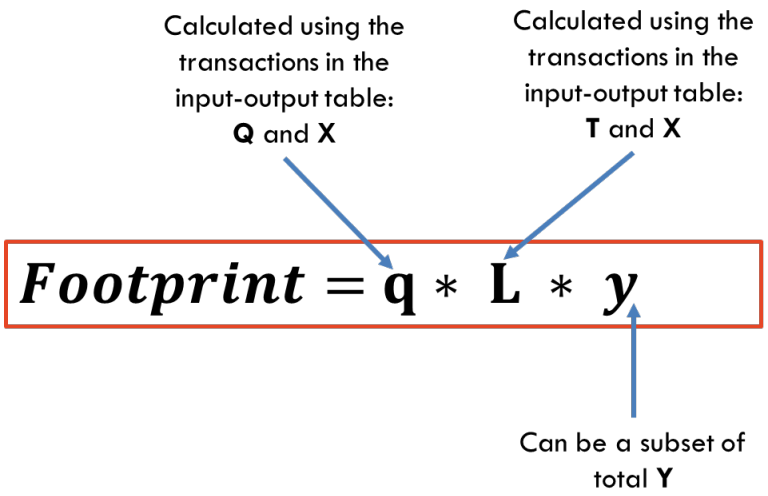
Examples:

y_{me} = My household expenditure → **q*L*** y_{me} = My household footprint

y_{USYD} = Sydney Uni's expenditure → **q*L*** y_{USYD} = Sydney Uni's footprint

y_{QANTAS} = QANTAS' expenditure → **q*L*** y_{QANTAS} = QANTAS' footprint

Calculating footprints for a particular entity



1. Create the entity's y expenditure vector, but make sure the values in this vector are not greater than the values in the relevant final demand vector in the input-output table.

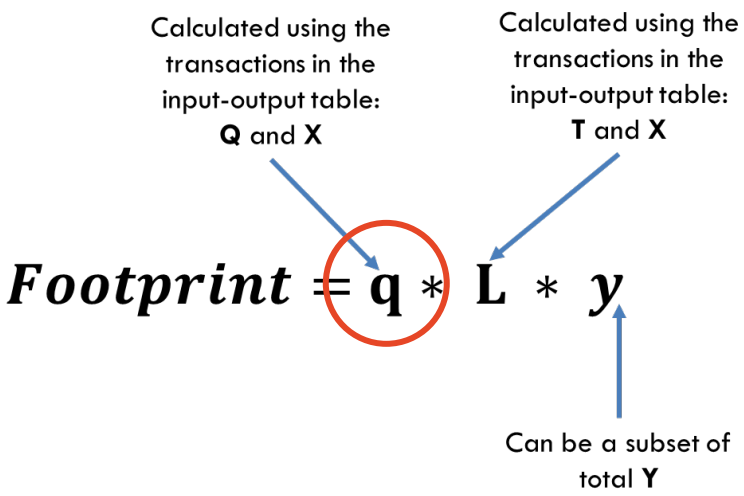
2. The expenditure vector will have the same number of rows as the **T** matrix, but will only have one column.

Calculating Scope 3 – part II

$$\text{Scope 3} = \mathbf{q} * \mathbf{L} * \boldsymbol{y}_{\text{ent}} - \text{Scope 2} \; (- \text{Scope 1 sometimes*})$$

* If the entity's own scope 1 emissions are significant enough and/or they make a significant contribution to the economic system, they will also appear in the entity's supply chains, since other sectors are dependent on them. This needs to be managed on a case-by-case basis.

Reporting for other indicators



$\mathbf{q}$ = Number of people in conditions of modern slavery per $1 of total output → Modern slavery footprint

Potential source for Q: ILO

$\mathbf{q}$ = Measure of extinction risk per $1 of total output (e.g. nSTAR) → Extinction-risk footprint

Potential source for Q: IUCN Red List

$\mathbf{q}$ = Land use area per $1 of total output → Land use footprint

Potential source for Q: FAO

$\mathbf{q}$ = Water use per $1 of total output → Water use footprint

Potential source for Q: FAO

Real life example

Dr Vivienne Reiner
Integrated Sustainability Analysis (ISA)

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Footprint applications Tutorial

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Tutorial Activity
Calculation of individual footprints

In week 5, we constructed a satellite block for a 10-sector input-output matrix. The satellite block had four indicators – CO_2 emissions, energy, water and nitrogen. In week 6, you calculated the consumption footprints using this matrix.

($100million)	Sheep, Grain, Beef and Dairy Cattle	Poultry and Other Livestock	Coal Mining	Beer Manufacturing	Textile Manufacturing	Chemical Manufacturing	Retail Trade	Technical Services	Banking	Defence	Households	Government	Exports
Sheep, Grain, Beef and Dair Cattle	40	30	1	40	2	4	1	1	0	0	3	10	72
Poultry and Other Livestock	35	25	0	2	10	10	5	1	0	0	20	5	48
Coal Mining	2	1	25	0	0	10	0	15	0	0	0	12	125
Beer Manufacturing	1	2	5	38	0	2	10	0	0	0	14	10	59
Textile Manufacturing	10	5	0	5	25	12	3	1	0	5	10	6	28
Chemical Manufacturing	22	25	42	8	9	24	1	2	0	32	5	15	15
Retail Trade	12	8	4	3	12	4	24	28	21	8	32	8	12
Technical Services	24	25	19	12	20	30	8	20	48	50	15	20	7
Banking	8	7	25	6	4	12	15	25	30	50	28	40	44
Defence	0	0	0	0	0	0	0	0	0	12	0	146	100
Households	12	3	15	5	6	10	55	85	63	10			
Government	10	9	13	4	5	10	15	12	20	52			
Imports	28	21	41	18	17	72	39	108	112	39			

Use this input-output table matrix to complete the following:

- **Construct a new final demand column (you can call it "My shopping basket").**
  Add data in the new column to reflect your buying choices.

- **Calculate the footprint for your personalised shopping basket.**
  Use the equation qLy to calculate the consumption footprint for your shopping basket.

- **Calculate the footprint for your consumption of goods from a specific sector (e.g. Agriculture).**
  To calculate the footprint for a specific sector, do the following a) replicate your shopping basket vector; b) make all values zeros, except for agriculture; and c) calculate the footprint using qLy, make sure you use the new y from part b).

Homework for Week 8

Homework week 8 (Due: October 28, 2026)

Use your 14-sector table to complete the following:

1. **Construct a new final demand column (you can call it "My shopping basket").**

Add data in the new column to reflect your buying choices against the sectors in your IO table (these can be hypothetical).

2. **Calculate the footprint (for any one indicator) for your personalised shopping basket.**

Use the equation **qLy** to calculate the consumption footprint for your shopping basket.

EXPLANATION: Suppose you have data on consumption for an individual in dollars and the input-output table is in million dollars. What unit conversions will you undertake for calculating footprints? What process would you follow? What equations would you use to calculate the footprint associated with that individual's consumption? (100 words)

PHYS5033 organisation – assessment tasks

Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the homework steps, make sure you provide an explicit answer for that question in the Excel file.
- Always include labels and units whenever you present results and make sure that your workings are included.
- You will be provided with worked examples and tutorial answers. These provide you with examples of how the calculations work – they do not provide you with a template for the homework. If you just replicate these, you will not be answering the steps expected in the homework.
- You should be able to answer the questions posed in the 'Explanation' part of the homework each week using the content provided in the lectures and in the background reading material. There is no need to use other sources to answer these questions, and doing so often results in students providing irrelevant or incorrect answers. If you do use other sources of information, these sources must be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

Key takeaways

- Input-output analysis can be applied to a range of different sectors and indicators

- Reporting Scope 3 emissions as defined by the GHG Protocol is not quite the same as using the consumption-based footprint that input-output analysis provides us (but you can work it out!)

More to come:

- Analysis techniques using input-output analysis – production layer decomposition (PLD) and Structural Path Analysis (SPA)